

Mizan ميزان

An agent that owns contract review, and knows when not to decide

Reads Arabic, English and bilingual contracts, typed or scanned. Checks every clause against the buyer’s playbook. Approves, negotiates with a redline, or escalates to a lawyer with the exact question.

Mohammad Albeladi · AI trainer, SDAIA Academy and KAUST · 30–31 October 2026

Problem statement

Legal reads every contract by hand, and most of them are routine

- Supplier contracts arrive in Arabic, English or both, often as scans.
- Lawyers spend their time confirming that routine contracts are routine.
- The risky ones look like the routine ones: an uncapped liability, a monthly late charge, personal data sent abroad.

The expensive mistake is a confident approval.

Automating review is easy. Automating it so that nothing risky is signed without a lawyer is the problem we set out to solve, and the number we measure first. Most tools leave that decision to the model.

Proposed solution

Every contract ends in one of three routes, decided by code

Approve

Meets the playbook, and the evidence is on record: every clause read, the cap on our liability quoted. No one touches it.

Negotiate

Major deviations. Mizan writes the redline and the message to the counterparty, citing the contract’s own article numbers.

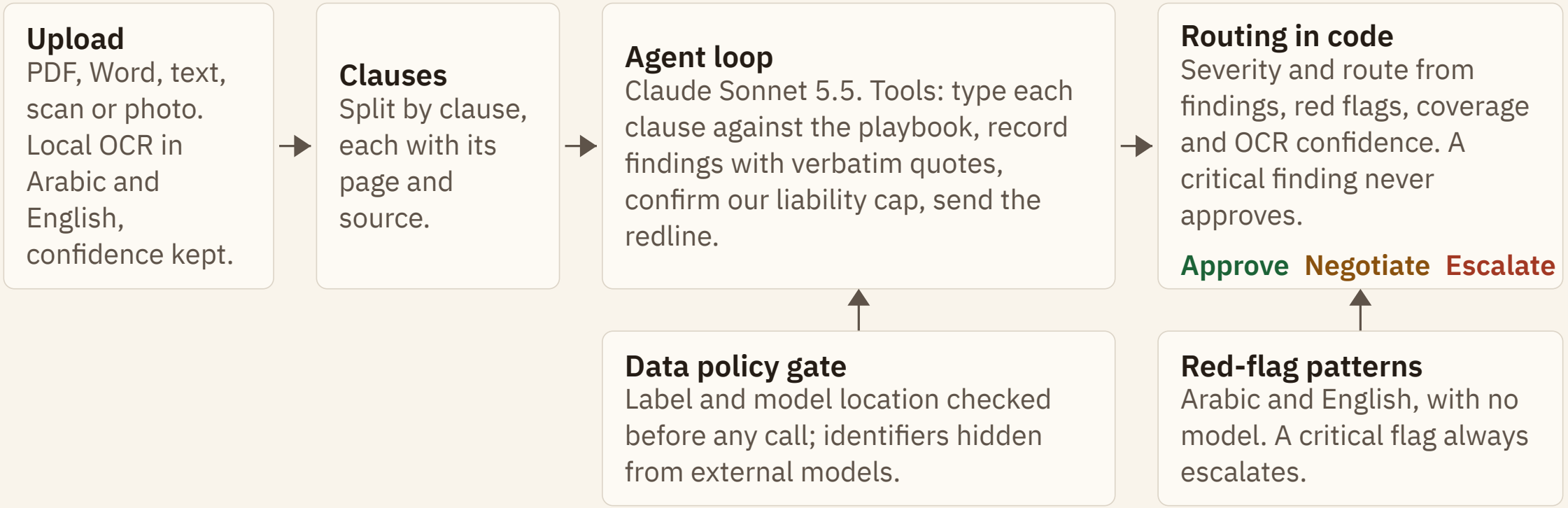
Escalate

Critical risk, a missing clause, an injected instruction, a doubtful scan. A lawyer gets it with the exact question.

The model reads, types clauses and records findings with verbatim quotes. It never chooses the route. Only escalations reach the lawyers’ inbox.

Technical approach

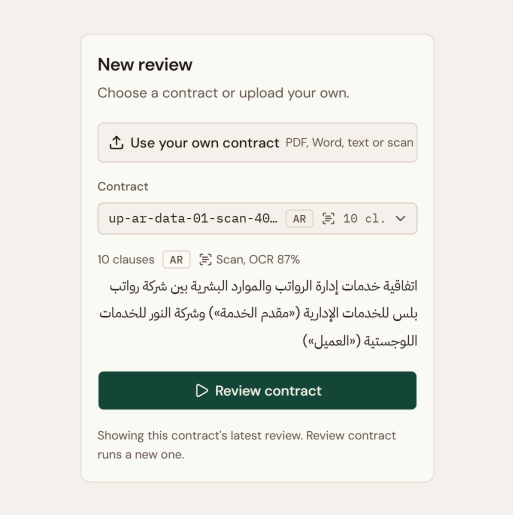
The model reads; code decides

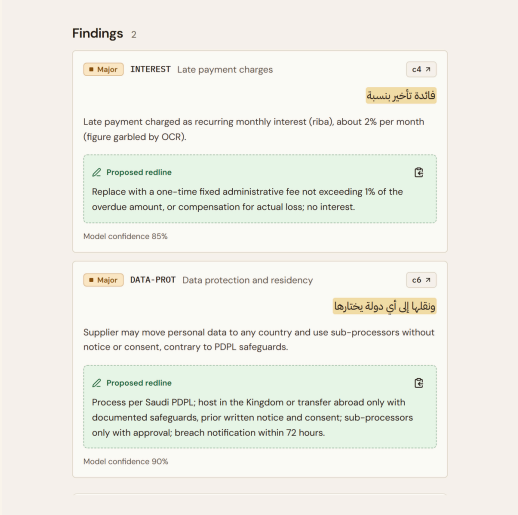


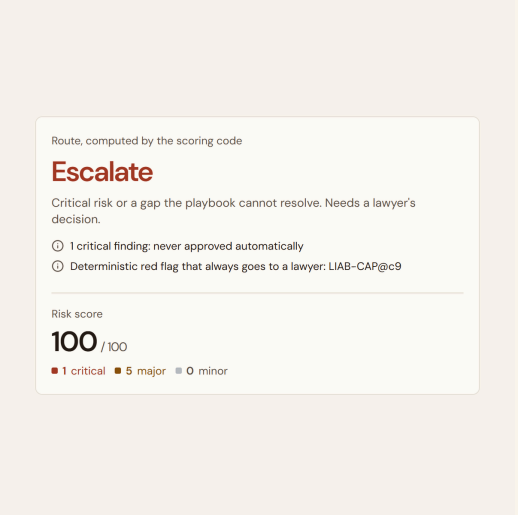
Models: Claude Sonnet 5.5, effort medium, through the Anthropic SDK; or a local model through an OpenAI-compatible server (qwen3:14b measured). Every run has a turn cap, a USD budget and a JSONL trace. Python 3.12, FastAPI, React with Tailwind and shadcn/ui, Tesseract OCR, Typst PDFs, one Docker image.

Implementation

Upload, check every clause, route







1 Bring any contract

PDF, Word, text, scan or photo. Scans are read by local OCR and the confidence is kept.

2 Every clause checked

Findings quote the contract word for word, with the playbook's wording as a redline.

3 Code routes it

Severity and route come from code. A critical finding never approves.

Also built: a lawyers' inbox that holds only escalations, a batch queue with an auto-send policy, redline and memo PDFs typeset from data, a Proof screen with every result, Mizan's own sign-in, and a guided demo that replays recorded runs.

Testing and results

59 contracts, 27 real. No risky one approved.

Method, for every set: pick the contracts by a rule, write the answer key (reviewers who never saw Mizan's output), commit it to git, run once, grade by code. Blind sets are reported as they came out.

Test set	What it is	Routes right	Risky approved	Recall	Per contract
Main, 4 runs	12 we wrote and tuned on; same result every run	48/48	**0/36**	100%	$0.04
Blind	12 by a separate agent, new to the system	12/12	**0/8**	92%	$0.05
Blind, batch 2	8 new industries	8/8	**0/6**	100%	$0.05
Real	SEC filings, ToS, Saudi government (Arabic, to 101 pages)	14/14	**0/14**	73%	$0.23
Real, batch 2	12 SEC filings picked by a rule, key from CUAD annotations	11/12	**0/12**	66%	$0.18

Same results with names hidden (12/12, 0/9) and under another company's playbook (8/8, 0/5). On real contracts it catches liability, interest, indemnity and non-compete issues and misses many termination and payment-term ones. Sonnet 5.5.

Innovation and value

Approval needs evidence, not just no findings

Code decides.
A critical finding never approves or negotiates, whatever the model writes.

Coverage, proven.
Every page read, every clause typed, the words capping our liability quoted and verified.

Two detectors.
Arabic and English red-flag patterns run without the model. A critical flag cannot be argued away.

Doubt escalates.
Low OCR confidence, an unclear clause, an instruction aimed at the reviewer, an unfinished run.

What it is worth

- Lawyers see only what needs them; routine contracts are approved with the evidence on record.
- About 5 cents and 17 seconds for a typical contract.
- Arabic and bilingual contracts, and scans read by local OCR with the confidence kept.
- Another buyer’s rules load from a file; lawyers’ decisions become proposed amendments a person accepts.

Innovation and value · data policy

Confidential contracts never leave

Label	External model (API outside the network)	Model in the network (local or HUMAIN-hosted)
Public templates	Reviewed as is.	Reviewed as is.
Internal names, IBANs	Identifiers become placeholders before sending, and are put back in everything people read and send out.	Reviewed as is.
Restricted confidential	No model call at all. The contract goes straight to a lawyer, with the reason.	Reviewed, but never approved or sent automatically. A lawyer signs off.

Mizan suggests the label from the text before any model sees it; making it looser is recorded as a person's decision. Hidden: names, emails, phones, IBANs, ID, CR, VAT.

Enforced twice in code: before the run starts and in the tool that serves contract text. Every review and memo PDF records the label, where the model ran and what was hidden.

Review quality with identifiers hidden: **12/12 routes right and 0/9 risky contracts approved on the main suite, the same as with names shown**

Challenges and future improvements

The first real run approved 5 of 14. One rule fixed it.

5/14

wrongly approved on the blind first run

0/14

after one general rule

On Saudi buyer-side paper the agent read the **supplier's** uncapped liability as the liability clause, so nothing capped **ours**. Approval now needs the words that cap our liability, quoted and verified by code; without them the contract escalates. The earlier suites were re-run afterwards and held, and 12 new real contracts it had never seen, run once blind, gave 0/12.

Limits we state rather than hide

- Playbook rules are illustrative, not legal advice; most labels are ours, 15 come from outside annotators.
- On real contracts recall is 66-73%: the route is safe, the issue list is incomplete.
- No real contract in our sets should be approved, so approving a real compliant contract is untested.
- Red-flag patterns catch 1/13 blind; a local model still approved 2/8, so local mode needs a lawyer on every contract.
- The OCR gate is calibrated on three sample scans; handwriting, stamps and signatures are not read.

Next

- Lawyer-validated labels, including risky Arabic contracts, where no public labelled source exists.
- Real compliant contracts in the suite, so approvals are exercised and not only refused.
- Better recall on termination and payment terms, measured on a fresh blind batch.

Conclusion and demo

The model reads. Code decides. Anything uncertain goes to a lawyer.

- On 47 contracts it was not tuned on, 27 of them real and public: 45 of 46 labelled routes right and 0 of 40 risky contracts approved. The main suite gave the same result in 4 runs.
- Arabic, English and bilingual, typed or scanned, up to 101 pages read by OCR.
- Another buyer’s rules from a file, a local model option, and confidential contracts that never leave.
- 118 tests with no API calls, CI on a fresh clone, every number on these slides reproducible from the repo and shown on the Proof screen.

Demo

`Demo video link`

About four minutes: upload, the findings with their quotes, the route, the lawyers’ inbox and Proof. The guided demo replays four recorded runs, labelled Replay, with no key and no network.